What makes an approach to modelling language 'symbolic'?

Ann Copestake

July 2026

Why I'd like to talk about this

- Immediate reason: Gemma Boleda 'LLMs as a synthesis between symbolic and distributed approaches to language' EMNLP, Findings, 2025.
- My current interests relate to the history of computational linguistics and improving general / interdisciplinary understanding of the current technologies.
- Older CL literature (50s and 60s) uses *symbolic* in several ways, but not in the current 'symbolic' vs 'non-symbolic' use.
- I feel it may be worth clarifying and disentangling as a way of improving the precision of discussion.
- Aim here is to start with a discussion via examples, and then (if there's interest), discuss some of the ideas in various literatures.

https://commons.wikimedia.org/wiki/File:MONIAC_computer.jpg

MONIAC: Monetary National Income Analogue Computer (Phillips, c1950)

Uses coloured water to model the 'national accounts' (money flow in the economy)

Models combination of relationships such as $Y \propto M1$ (Y is national income)

Can we agree that this is **non-symbolic**?

Meaning representation: symbolic and non-symbolic

Logical representation:

$$\forall x[\text{rabbit\_n\_1}'(x) \implies \text{animal\_n\_1}'(x)]$$

Embeddings:

	dim1	dim2	dim3	dim4	dim5	dim6	...
rabbit	0.08623	0.00004	0.01613	0.00023	0.01023	0.51623	...
sheep	0.00625	0.00003	0.01623	0.00001	0.20944	0.42277	...

Early distributions (e.g., Harper, 1965; Spärck Jones, 1967)

Extract adjective-noun pairs from some text (actual experiments on Russian):

white rabbit	hungry rabbit	beautiful rabbit	fat rabbit		
white sheep	hungry sheep	beautiful sheep	fat sheep	crazy sheep	
white chair					new chair
		beautiful idea		crazy idea	new idea

Represented with Boolean values in vectors (using 0 and 1 but could be F and T):

	white	hungry	beautiful	fat	crazy	new
rabbit	1	1	1	1	0	0
sheep	1	1	1	1	1	0
chair	1	0	0	0	0	1
idea	0	0	1	0	1	1

Early distributions, continued

Is this symbolic?

If not, why not?

If it is symbolic, at what point does distributional work become non-symbolic?

Note:

- This representation allows a continuous measure of similarity via Jaccard.
- If Boolean-valued, the **metric space** is discrete rather than continuous. (Also true if integer-valued.)
- We can talk about distributions in terms of logic and denotation (CLRU early work, see also Herbelot and Copestake, 2021).
- word2vec embeddings are formally identical to ordinary count distributions with dimensionality reduction

Symbolic grammars plus stochastic ranking/probabilities

- Probabilities purely as surrogates for world knowledge and context?
- Probabilities model some linguistic aspects? (Various possible examples.)
- Note: when modelling a corpus with many different speakers, we might get probabilities even if they aren't relevant to individuals.

Some further possible issues

Probably not relevant to the definition:

- Human-written vs machine learning (though don't expect to be able to manually create a non-symbolic account).
- Underlying system or hardware: e.g., could create a logic-based system out of fluids.
- Monotonicity: non-monotonic logics are symbolic.

Worth thinking about?

- Interpretability: but symbolic system with lots of rules vs MONIAC?
- Complexity of categories: e.g., degrees of nouniness or verbiness in non-symbolic system, vs more categories in symbolic system (capturing generalizations).
- For grammars etc, degrees of reliance on symbolic system vs probabilities.
- Degree of granularity (*density* in the literature): not useful to talk about systems with lots of big integers as symbolic?
- Underlying formalism: can a system without a formalism be symbolic? (Non-symbolic systems can have underlying equations, however.)
- Soft categories: **near-symbolic**?

Word senses, very informally

1. standing for something else, possibly poetically
2. relating to words etc in natural language or an artificial language
3. relating to formal / mathematical symbols / product symbols etc
4. *symbolic logic*

Some traditions

1. Peirce: *symbol* contrasted with *icon* and *index* (simplifying . . .)
2. symbolic logic: *symbolic* seems to have been used just because of the squiggles . . .
3. Nelson Goodman (1969): *Languages of Art*
   https://homepages.inf.ed.ac.uk/jlee/mac-john/VRI-Lee-final.html
   MONIAC **symbolises** but not a **symbolic notation** (I think!)
4. cognitive science: especially in discussions about human uniqueness, properties of mental representations that persist and connect to language
   Platner (2019) 'What is Symbolic Cognition?'

Metaphors

- Symbolic approach (symbolic notation): associated with correctness but also inflexibility. Extends the traditional understanding of computers, which was flawed, even without stochastic processes (e.g., distributed systems).
- Non-symbolic approaches: mainstream is to push analogy with humans, but that's both wrong and deeply unimaginative.
    - Early approaches to analogue computers provide alternative ideas (e.g., Phillips, Gordon Pask and colleagues).